2. Diadelphous: Filaments of stamens connate into two bundles. Example: *Fabaceae (pea)* and *Clitoria*.

3. Polyadelphous: Filaments connate into many bundles. Example: *Citrus*, *Bombax*

b. Syngenesious: Anthers connate, filaments free. Example: Asteraceae.

c. Synandrous: Filaments and anthers are completely fused. Example: *Coccinea*.

2. Adnation: Refers to the fusion of stamens with other floral parts. **Epipetalous** : Stamens are adnate to petals .Example: brinjal, *Datura*.

a. Episepalous: stamens are adnate to sepals. Example: *Grevillea* (Silver oak)

b. Epitepalous (**epiphyllous**): stamens are adnate to tepals. Example: *Asparagus*.

c. Gynostegium:Connation product of stamens and stigma is called **gynostegium**. Example: *Calotropis* and Orchidaceae.

d. Pollinium: Pollen grains are fused together as a single mass Example: *Calotropis*

4.4.2 Arrangement of stamens relate to length of stamens:

1. **Didynamous:** Four stamens of which two with long filaments and two with short filaments. Example: *Ocimum*

2. **Tetradynamous**: Six stamens of which four with long filaments and two with short filaments. Example: *Brassica*.

3. **Heterostemonous**: stamens are of different lengths in the same flower. Example: *Cassia*.

4.4.3 Anther types

1. Monothecal: One lobe with two microsporangia. They are kidney shaped in a cross section. Example: Malvaceae



Figure 4.19: (a) Monothecal



Figure 4.19: (b) Dithecal

2. Dithecal: It is a typical type,having two lobes with four microsporangia.They are butterfly shaped in cross section. Example: Solanaceae.

4.4.4 Anther attachment

1. Basifixed:(Innate) Base of anther is attached to the tip of filament. Example: *Datura*.

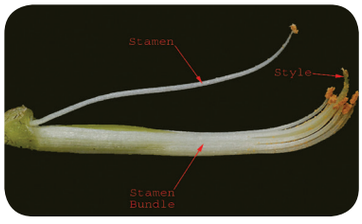
2. Dorsifixed: Apex of filament is attached to the dorsal side of the anther. Example: *Hibiscus*.

3. Versatile: Filament is attached to the anther at midpoint. Example: Grasses.

4. Adnate: Filament is continued from the base to the apex of anther. Example: *Nelumbo*



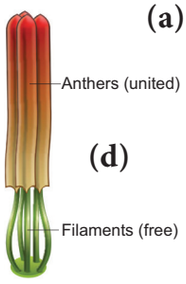
(a)



(b)



(c)



(d)

Figure 4.17:
(a) Monadelphous,
(b) Diadelphous,
(c) Polyadelphous
(d) Syngenesious

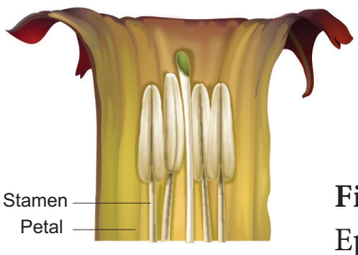


Figure 4.18: Epipetalous

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